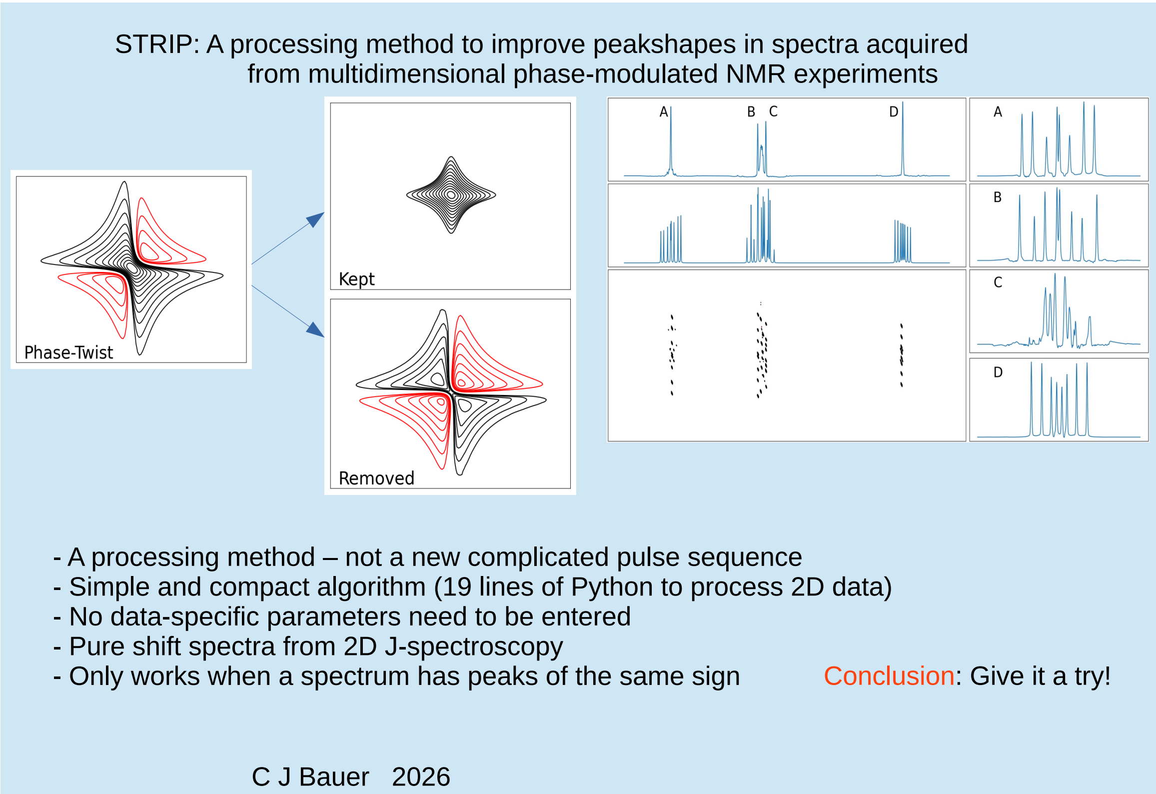
Graphical Abstract

STRIP: A processing method to improve peakshapes in spectra acquired from multidimensional phase-modulated NMR experiments

Christopher John Bauer



Highlights

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- A new processing method to reduce the double dispersive components of spectra containing phase-twist peaks
- Can be applied to a 2D homonuclear J-spectrum to get clean multiplets and a shift-only spectrum
- Algorithm is very compact and easy to understand
- No data-specific parameters need to be entered
- The $\mathcal{B}$ transform is introduced

STRIP: A processing method to improve peakshapes in spectra acquired from multidimensional phase-modulated NMR experiments

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Abstract

An issue that was identified in the early days of two-dimensional NMR was that peaks observed from phase-modulated data had unpleasant *phase-twist* shapes. These are peaks containing a mixture of double absorption and dispersion components. These cannot be separated by simple phase correction. In this paper a new processing method is introduced, STRIP, that can reduce the size of the double dispersion components in such peaks. The algorithm is unable to produce a totally pure absorption mode spectrum, but manages to reduce the dispersive components by such an extent that the problems associated with them are greatly alleviated. The algorithm behind the processing method is compact and easy to understand. The algorithm makes use of the $\mathcal{B}$ transform which is introduced in this paper. A limitation of the method is that it should only be applied to spectra where all the required absorption peaks are of the same sign.

Keywords: STRIP, $\mathcal{B}$ Transform, Phase-Twist, Homonuclear J-Spectroscopy, Pure Shift

1. Background

The theory regarding shapes of multidimensional NMR peaks has been explained in detail elsewhere[1]. The equations presented there have reduced below to the simplest representation that can still explain the issues that are faced and the solution that is being provided. Beginning with the one-dimensional NMR experiment, the result of applying a complex Fourier transform to a free induction decay produces an array of complex points. The real and imaginary components of this array each provide a representation of the spectrum. Conversion between these representations is possible by applying an appropriate $\pi/2$ phase shift. The NMR signals appearing in the spectrum will have intensities and lineshapes depending on a multitude of factors such as the relative sensitivity, concentration and relaxation properties of the nuclei generating the signal and instrumental factors such as the strength and homogeneity of the magnetic field and relative phases of the transmitter pulse and the quadrature receiver channels. The new processing method presented in this paper can be applied to spectra containing many signals, however, it is simple and sufficient to describe the method using just a single signal. The complex one-dimensional spectrum from such a single signal will simply be written as

$$\text{One-dimensional spectrum} = A + iD \tag{1}$$

where all the above factors that would affect the intensity of the signal have been ignored. These factors have no relevance to the argument that is being presented so it is convenient to assume unit intensity. Since there is just a single signal there is no need for labels identifying the nucleus or frequency within the spectrum. A and D represent components with absorption and dispersion lineshapes respectively. It has been assumed that if necessary a phase adjustment is applied to the spectrum to ensure that A, the real part of the spectrum, has an absorption lineshape and that D, the imaginary part, has a dispersion lineshape. The assumption that a phase correction has been applied to one-dimensional spectra to ensure that A and D have the correct lineshapes is used throughout. For example, when considering two-dimensional NMR data it is required that this phase correction is applied to all increments prior to the second Fourier transformation step. As well as being a required step, the application of this phase correction

Email address: chris_bauer_nmr@mankbox.duckdns.org (Christopher John Bauer)
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simplifies the description of the properties of the two-dimensional spectrum. The phase modulation and decay of the 1D spectra acquired in a 2D NMR experiment can be expressed as follows

$$\text{Phase-modulated 2D data} = \exp(i\omega t_1)\text{decay}(t_1)[A_2 + iD_2] \tag{2}$$

Note that subscripts have now been added to indicate the dimension in the 2D data. The second Fourier transform with respect to t_1 gives us a spectrum of the form

$$\begin{aligned}\text{Two-dimensional spectrum} &= [A_1 + iD_1][A_2 + iD_2] \\ &= A_1A_2 - D_1D_2 + i[A_1D_2 + A_2D_1]\end{aligned} \tag{3}$$

The real part of the spectrum consists of two components A_1A_2 and $-D_1D_2$. In addition to the desirable pure double absorption component, A_1A_2, there is also the undesirable double dispersive $-D_1D_2$ component. It is not possible to apply a simple *phase correction*, that is taking a particular combination of real and imaginary parts of the spectrum, in order to leave just the pure absorption A_1A_2 spectrum. The apparent inseparability of the absorption and dispersion components has meant that historically both of these components must be displayed together. The combined lineshape, which has both positive and negative excursions, is sometimes known as a *phase-twist* (or *phase-twisted*) lineshape.

The component parts of a phase-twist peak are shown in the top line of figure 1. The pure absorption component on the left is much more compact than the phase-twist peak. It is clear that the resolving power of a two-dimensional NMR experiment would be greatly enhanced if the dispersion components could be removed and only absorption components remained. The negative excursions of the phase-twist peaks are problematic too. Potentially they could hide a neighbouring low intensity peak. In order to avoid the negative excursions of the phase-twist lineshape a frequently adopted solution is to display the spectrum in absolute value mode. The middle line of figure 1 shows that such a display mode fixes the problem with negative excursions. However, dispersive components in the real and imaginary parts of the spectrum combine to increase the length of *tails* emanating from the centre of the peak in the direction of the compass points. It would require extremely aggressive resolution enhancement, such as a pseudo Gaussian echo[2], to address the problems caused by these tails. In order to avoid phase-twist peakshapes experiments are typically designed to acquire data such that it appears to have been amplitude modulated [3, 4, 5]. Some experiments, typically those where there is no coherence transfer[1], cannot be adapted to produce amplitude modulated data and the problem of phase-twist peaks remains.

This paper presents STRIP, a new processing method which seeks to remove the double dispersion component from phase-twist peaks. The naming of the method derives from it being a <u>S</u>imple <u>T</u>ail-<u>R</u>educing <u>I</u>terative <u>P</u>rocedure. An example of the application of STRIP is shown in the bottom line of figure 1. This is not the first time a data processing method has been applied in search of such a solution [6, 7, 8, 9, 10]. STRIP has no requirement on the type of absorption component that is being resolved. It could be Lorentzian, Gaussian or any other. All that matters is that in each dimension the dispersion component is related to the absorption component by a $\pi/2$ phase shift. A key step in the STRIP algorithm involves a transformation of the data. Before describing the algorithm in detail the new transform will be explained.

2. The $\mathcal{B}$ Transform

We define a transform $\mathcal{B}$ (and its inverse $\mathcal{B}^{-1}$) that interconverts between A_1A_2 and $-D_1D_2$, such that

$$\begin{aligned}\mathcal{B}(A_1A_2) &= -D_1D_2 \\ \mathcal{B}(D_1D_2) &= -A_1A_2 \\ \mathcal{B}^{-1}(A_1A_2) &= -D_1D_2 \\ \mathcal{B}^{-1}(D_1D_2) &= -A_1A_2\end{aligned} \tag{4}$$

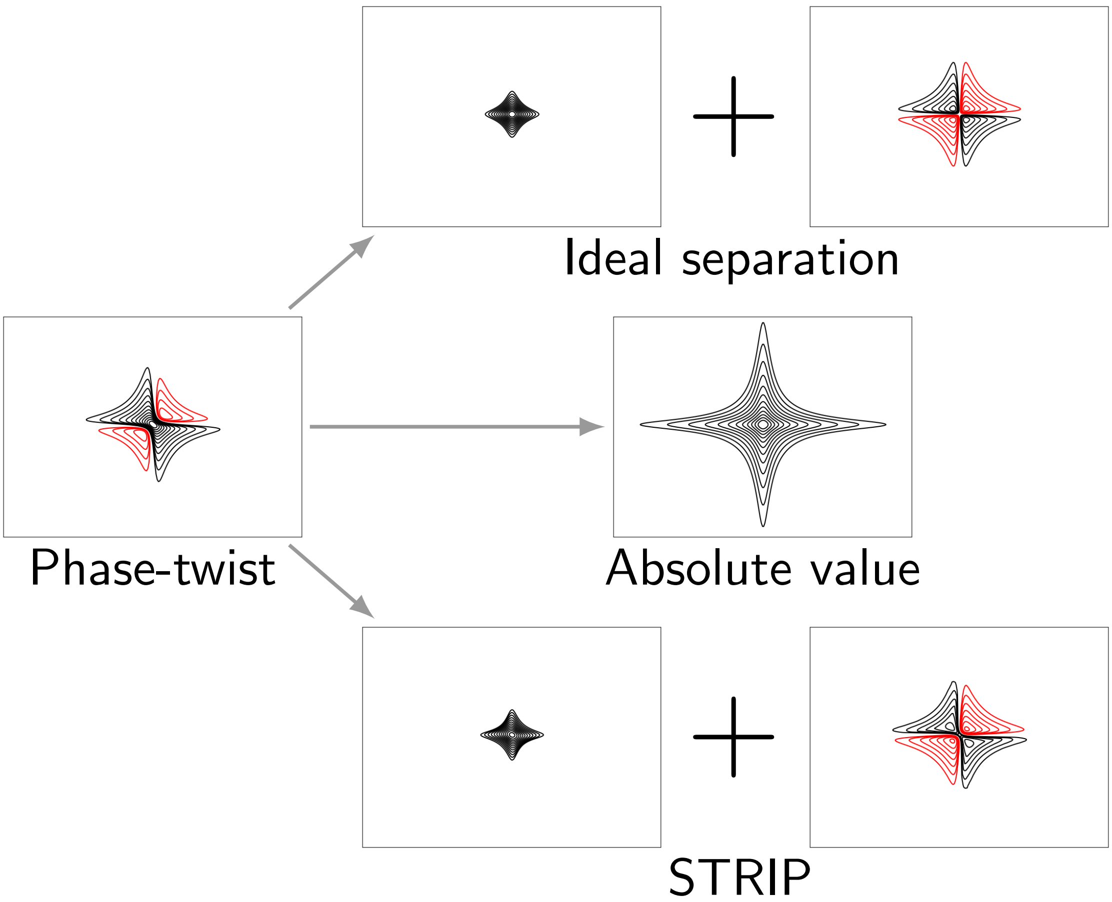

Figure 1: The spectrum on the left shows the phase-twist peak that appears as the real part of the spectrum arising from a two-dimension complex Fourier transform of a phase-modulated signal. Such a peak is the sum of an absorption signal and a dispersion signal. The top line in the figure shows what these components would look like if it were possible to separate them. The middle line shows the absolute value spectrum that can be calculated by taking the square root of the sum of squares of the real and imaginary parts of the spectrum. The bottom line shows the result of applying the STRIP algorithm to separate the phase-twist peak into absorptive and dispersive components.

It follows that $\mathcal{B}$ and its inverse are identical operations. Furthermore if the transform is applied to a spectrum containing a phase-twist peak we get the following

$$\begin{aligned}\mathcal{B}(\text{phase-twist peak}) &= \mathcal{B}(A_1A_2 - D_1D_2)\\ &= -D_1D_2 + A_1A_2\\ &= \text{phase-twist peak}\end{aligned} \tag{5}$$

It follows that applying the $\mathcal{B}$ transform to a 2D spectrum containing only phase-twist peaks leaves the spectrum unchanged. In practice the data sets that we handle contain noise as well as peaks. It is interesting to consider the effect of the $\mathcal{B}$ transform on noise. It would be tempting, but incorrect, to assume that a $\sqrt{2}$ improvement in signal-to-noise ratio could be achieved by adding together a spectrum and the result of a $\mathcal{B}$ transform on that spectrum. The incorrect assumption would be that peak intensities would double and that noise intensity would only increase by a factor of $\sqrt{2}$ if the noise in the spectrum and its transform were independent. However, that is not the case. There is no improvement in signal-to-noise ratio. This is because as well as the phase-twist peaks, the noise in the spectrum is also unmodified by the $\mathcal{B}$ transform.

It is important to clarify that this is due to a special property of the noise in the spectrum rather than a special property of the $\mathcal{B}$ transform. The noise in the spectrum produced by a two-dimensional Fourier transform on zero-filled time-domain noise has itself a phase-twist character and is not truly random. To summarize, we have now defined a transform that has no effect whatsoever on either the peaks or noise in our spectrum. It might for now seem somewhat useless, however, it is actually the key tool in the STRIP algorithm.

3. The STRIP Method

In a nutshell, STRIP takes a guess of the pure absorption spectrum and uses the $\mathcal{B}$ transform to convert this into a guess of the double dispersion spectrum that would need to be subtracted from the phase-twist peaks in the real input spectrum in order to leave peaks with double absorption shapes. The method is iterative. The subtraction does not produce the desired absorption spectrum from the initial guess, instead the output from a pass through the algorithm is used to generate the guess for the next iteration. Eventually a *convergence* point is reached where an additional iteration would provide no visible improvement to the spectrum.

Before describing the method in detail the requirements that the input spectrum must have are set out below.

Complex: The input spectrum must have real and imaginary components.

Zero Filled: The input spectrum must have been created by processing using zero filling such that the total size of time-domain data in each dimension was at least twice that of the acquired data.

Phased: Every effort has been made to ensure that there are pure phase-twist peaks in the real part of the input spectrum. Before the second Fourier transform, a phase correction has been applied to the data that matches that needed to phase a 1D spectrum (the first increment of the experiment). If the timing of the pulse sequence introduces a first-order phase error in F_1 this should also be corrected. Note that each peak requires a single value of phase adjustment to account for precession occurring during a delay introduced by a timing error. However, a first-order phase correction is a frequency-dependent adjustment. If a peak is broad, applying one will lead to a range of adjustments being applied across the peak instead of just a single adjustment for the peak. The phase gradient that is introduced across the peak manifests itself through baseline distortions. In general, first-order phase corrections should be avoided, for example by paying careful attention to timing to eliminate a delay or, if that is not possible, by sacrificing the first time-domain data point, correcting the acquisition times of subsequent data points, and applying a linear baseline correction in the frequency domain to imply a value for the missing first time-domain point.

Single Signed: The STRIP method assumes a target of a spectrum of positive absorption peaks in the real part of the spectrum. Attempts to improve spectra containing a mixture of positive and negative target peaks in the real part of the complex data will produce unpleasant looking peaks where negative peaks would have been. To process a spectrum containing all negative peaks the spectrum should be inverted before and after using the method.

3.1. Improving a Two-Dimensional Spectrum

A listing of an example implementation of STRIP to process 2D spectra in Python[11] is shown in listing 1. The listing shows that the algorithm makes use of functionality in the NumPy[12] and SciPy[13] Python modules. The use of these allows the algorithm to be coded in just 19 lines. The method is executed by calling the routine `STRIP`. This function takes as its input a two-dimensional NumPy array of type `complex` representing the spectrum. Optionally the function can be supplied a value overriding the default number of cycles to iterate.

The $\mathcal{B}$ transform converts A_1A_2 into $-D_1D_2$. This is done by applying a $\pi/2$ phase shift to both dimensions and negating the result. Applying a phase shift to an axis of a two-dimensional spectrum is different from applying a phase correction to a one-dimensional spectrum. The latter can simply be achieved by taking a different linear combination of real and imaginary components. Because the real and imaginary parts of a two-dimensional spectrum could each potentially contain any combination of the four possible peakshape components, taking a different linear combination of real and imaginary parts does not necessarily provide a simple phase correction to either dimension. The way to apply a $\pi/2$ phase correction to a single dimension of a two-dimensional spectrum is to use a Hilbert transform[1].

```python
import numpy as np
from scipy.signal import hilbert

def STRIP(input_spectrum, loops=15):
real_input_spectrum = input_spectrum.real
spectrum_guess = input_spectrum

for _k in range(loops):
initial_spectrum_guess = np.abs(spectrum_guess)
initial_dispersive_guess = B_transform(initial_spectrum_guess)
new_spectrum_guess = real_input_spectrum - initial_dispersive_guess
new_dispersive_guess = B_transform(new_spectrum_guess)
average_dispersive_guess = 0.5 * (new_dispersive_guess + initial_dispersive_guess)
spectrum_guess = real_input_spectrum - average_dispersive_guess
return spectrum_guess

def B_transform(spectrum):
h = hilbert(spectrum, axis = 0).imag
return -hilbert(h, axis = 1).imag
```

Listing 1: An implementation of the STRIP algorithm in Python

The $\mathcal{B}$ transform can be implemented by sequentially applying a Hilbert transform to every column and every row in the spectrum so that each dimension is phase-shifted by $\pi/2$ radians and then negating the result.

The listing includes an implementation of the $\mathcal{B}$ transform. This uses the `hilbert` function from the SciPy module to perform Hilbert transforms. The `hilbert` function takes as its input a real part of a spectrum and returns the real and imaginary part. Its implementation involves an inverse Fourier transform and manipulations that result in a halving of the length of the time-domain data. It is for this reason that the input spectrum needs to have been created with zero-filling in order to avoid loss of data. When passed a matrix the SciPy `hilbert` function performs a Hilbert transform on every spectrum along a given axis. There are potential baseline problems using the `hilbert` function. The usage of the $\mathcal{B}$ transform in STRIP is always to convert from absorption to dispersion. It does not matter that the `hilbert` function discards baseline information since a zero baseline is always required anyway. It follows that the implementation of the $\mathcal{B}$ transform shown in the listing is sufficient for its usage in STRIP.

The first step of the method is to make a guess as to what the target spectrum containing positive absorption peaks looks like. The initial guess used is the absolute value of the spectrum, which is the square root of the sum of squares of the real and imaginary components. This is the last occasion when the imaginary component of the input spectrum contributes to the processing. Beyond this point all matrices created in the method, including the final result, will be matrices of real values rather than complex. Obviously the peak shapes in the absolute value spectrum used as the initial guess are far from ideal, but they are going to have similar magnitudes and be in roughly the same place as the absorption peaks that are desired.

To make the explanation of STRIP as simple as possible consider calling it with an input spectrum containing just a single phase-twist peak.

$$\text{Real input spectrum} = A_1A_2 - D_1D_2 \tag{6}$$

A cartoon showing the various steps in the STRIP method is shown in Figure 2. The initial spectrum guess, a single broad absolute value peak, will consist of data points that are all real and positive. A representation of this absorption spectrum can written as a combination of the target absorption peak A_1A_2 and an error term. Since the spectrum is all positive it is convenient to consider the error term as being a superposition of many small absorption peaks. To keep this in mind the error term will be labelled as a_1a_2.

$$\text{Initial spectrum guess} = A_1A_2 + a_1a_2 \tag{7}$$

The next step in the method is to apply the $\mathcal{B}$ transform to the initial guess in order to estimate a guess of the double dispersion component contained in the phase-twist peak in the real input spectrum.

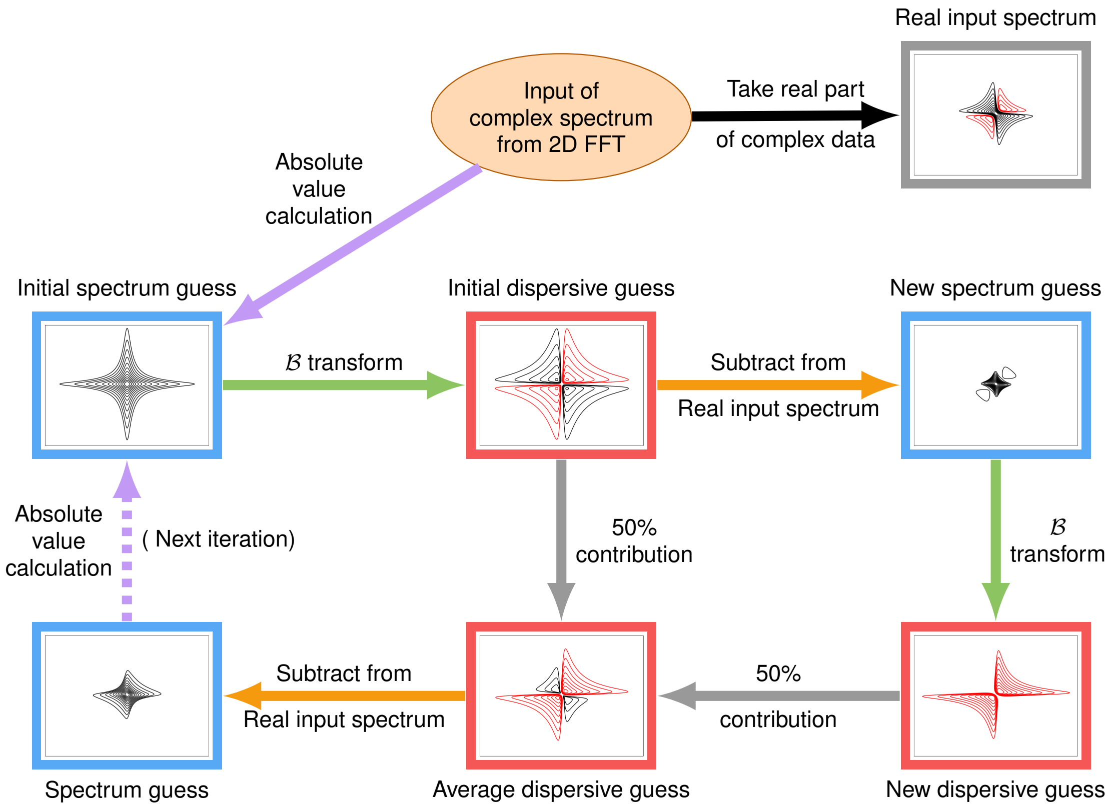

Figure 2: A cartoon showing the steps in the STRIP method. The peaks shown illustrate the first pass through the iterative STRIP algorithm where the input was the complex spectrum produced after two-dimensional Fourier transformation of a single phase modulated signal. The output is spectrum at the bottom left. Absorptive and dispersive spectra are shown in blue and red boxes respectively.

$$\begin{aligned}\text{Initial dispersive guess} &= \mathcal{B}(A_1A_2 + a_1a_2)\\ &= -D_1D_2 - d_1d_2\end{aligned} \tag{8}$$

where d_1d_2 is the error term transformed into a dispersive shape. The dispersive guess that has been calculated can now be subtracted from the real part of the input spectrum to refine the guess.

$$\begin{aligned}\text{New spectrum guess} &= \text{Real input spectrum} - \text{Initial dispersive guess}\\ &= A_1A_2 + d_1d_2\end{aligned} \tag{9}$$

Next the $\mathcal{B}$ transform is applied to this refined spectrum guess to create a new dispersive guess.

$$\begin{aligned}\text{New dispersive guess} &= \mathcal{B}(\text{New spectrum guess})\\ &= \mathcal{B}(A_1A_2 + d_1d_2)\\ &= -D_1D_2 - a_1a_2\end{aligned} \tag{10}$$

There are now two dispersive guesses. These are now averaged to create a refined dispersive guess.

$$\begin{aligned}\text{Average dispersive guess} &= 0.5(\text{Initial Dispersive guess} + \text{New dispersive guess})\\ &= -D_1D_2 - 0.5(a_1a_2 + d_1d_2)\end{aligned} \tag{11}$$

This refined dispersive guess is now subtracted from the real part of the input spectrum to create the output from a pass through the method. The STRIP method is iterative and this output will be used to generate the initial spectrum guess for the next pass through the algorithm.

$$\begin{aligned}\text{Spectrum guess} &= \text{Real input spectrum} - \text{Average dispersive guess} \\ &= A_1A_2 + 0.5(a_1a_2 + d_1d_2)\end{aligned} \tag{12}$$

It is useful to compare the output of a pass through the method with the input described in equation 7. The a_1a_2 component has been halved, and there is additional component $0.5d_1d_2$. The first step in the next pass of the method is to take the absolute value of this guess. The matrices we are dealing with by this stage in the processing only contain real values. Consequently taking the absolute value will simply convert negative values into positive values leaving a positive error term. It is possible to rewrite the initial spectrum guess for the next pass (in a way analogous to equation 7) as

$$\text{Initial spectrum guess} = A_1A_2 + a_1^1a_2^1 \tag{13}$$

where the numeric superscripts in the error component indicate the number of passes through the method. The maximum magnitude of these errors will be smaller on each pass through the method until convergence is reached. Iterative methods typically need some step that *pulls* each iteration towards convergence. In the STRIP algorithm the *pull* is provided by the step taking the absolute value of the spectrum guess. Convergence will happen when applying the absolute value function no longer has an effect on the spectrum guess. This would be the case when the spectrum is already entirely positive. In practice this convergence happens when the $d_1^nd_2^n$ part of the result is entirely covered by the A_1A_2 peak. Consequently the method does not converge on a pure double absorption solution, instead there is a slight distortion to the peaks that are produced due to some underlying residual components.

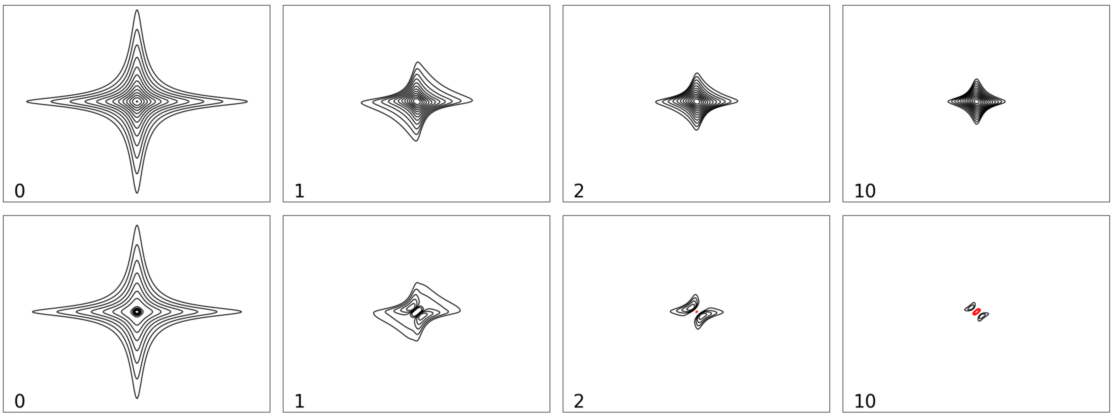

Figure 3: This shows the STRIP method in progress. The top line shows the application of STRIP on a single phase-twist peak. The number in the lower left corner shows the number of loops that have been executed. The peak on the top far left showing zero loops executed is the absolute value of the phase-twist peak that is used as the initial guess of the absorption peak. The lower line shows the difference between the peaks shown in the upper line and an idealized pure absorption peak.

Figure 3 shows how the STRIP algorithm progresses. After ten iterations the output is much more compact than the initial guess (which was the absolute value spectrum). The residual error, shown in the lower line of that figure, is greatly reduced indicating that the peakshape is relatively close to being an absorption peak. For a single isolated peak there will not be much more obvious visible improvement after 10 iterations. Low intensity errors which are not covered by peaks and are below the minimum contour level will be spread across the spectrum. Eventually after sufficient iterations have been performed only errors underneath peaks will persist. For this reason, where there is large dynamic range in a spectrum, for example where signals of interest coexist in a spectrum with a large phase-twist peak from a solvent, it is recommended to increase the number of iterations executed in STRIP beyond the default value of

```
def STRIP_quick_and_dirty(input_spectrum, loops=15):
real_input_spectrum = input_spectrum.real
spectrum_guess = np.abs(input_spectrum)

for _k in range(loops):
dispersive_guess = B_transform(spectrum_guess)
new_spectrum_guess = real_input_spectrum - dispersive_guess
new_spectrum_guess = np.abs(new_spectrum_guess)
spectrum_guess = 0.5 * (new_spectrum_guess + spectrum_guess)
return spectrum_guess
```

Listing 2: An implementation of the quick-and-dirty STRIP algorithm in Python

15. Note that the residual error pattern shown even after just 10 iterations is very close to the converged error pattern. That pattern infers a slight reduction in the maximum height of a peak that is processed by STRIP, but not its integral.

There is a simpler algorithm that averages spectrum guesses rather than dispersive signal guesses, see listing 2. The advantages of this method are that there is only one transform per iteration and that taking the absolute value happens before averaging. These factors ensure that the method executes faster and may even converge closer to the desired pure absorption result. Unfortunately the method also increases the intensity of underlying noise and so is not recommended unless sensitivity is not a problem.

3.2. Improving a Three-Dimensional Spectrum

The extension to higher dimensions is possible. Consider the peakshapes in a three-dimensional spectrum acquired using phase modulation.

$$\begin{aligned}\text{Three-dimensional spectrum} &= [A_1 + iD_1][A_2 + iD_2][A_3 + iD_3] \\ &= A_1A_2A_3 - A_3D_1D_2 - A_1D_2D_3 - A_2D_1D_3 \\ &\quad + i(A_1A_2D_3 + A_1A_3D_2 + A_2A_3D_1 - D_1D_2D_3)\end{aligned} \tag{14}$$

Note that the real part of the spectrum contains a pure absorption component and three components containing two dispersive D factors. In general, the real part of a spectrum with any number of dimensions always contains terms with an even number of D factors. Improving a three-dimensional spectrum using a method analogous to that presented in section 3.1 will involve removing the 3 terms containing two dispersive D factors. The initial spectrum guess will be the absolute value spectrum from the complex spectrum. The initial dispersive guess will involve applying a $\mathcal{B}$ transform along each of the three axes of the initial spectrum guess and summing to create an estimate of the term, $-A_3D_1D_2 - A_1D_2D_3 - A_2D_1D_3$, that it is desired to eliminate. Creating a new spectrum guess, new dispersive guess, average dispersive guess and finally a spectrum guess follow similar logic to that presented earlier.

4. Simulations

Simulations were performed on a PC running Linux containing an 8-core AMD 5800X processor and 16GB of memory. Phase-modulated time-domain data was generated and Fourier transformed with Python scripts using NumPy's multidimensional array functionality. Figures were plotted using the Matplotlib[14] Python module. The J-Spectrum time-domain data was generated using the GAMMA C++ library[15]. All the spectra shown in this paper were improved with STRIP using the Python script shown in listing 1. The number of iterations to execute was left at the default value of 15. Without any optimization a spectrum of size 2048x8192 took 33.2 seconds to process. Most of this time was spent within the `hilbert` function provided by the SciPy module. It is possible to enable parallelization by calling `scipy.fft.set_workers` to set the number of worker threads. On this system setting the number of workers to 6 reduced the execution time from 33.2 down to 12.8 seconds. The PC also contained a graphics card providing an Nvidia GeForce RTX 2060 GPU and 6GB of memory. This graphics card was released as a mid-range model seven years ago. The Python module CuPy[16] is closely related to NumPy and SciPy and allows execution of STRIP on the Nvidia GPU rather than the PC's CPU. Using CuPy reduced the execution time for STRIP on the spectrum of size 2048x8192 to just 2.6 seconds.

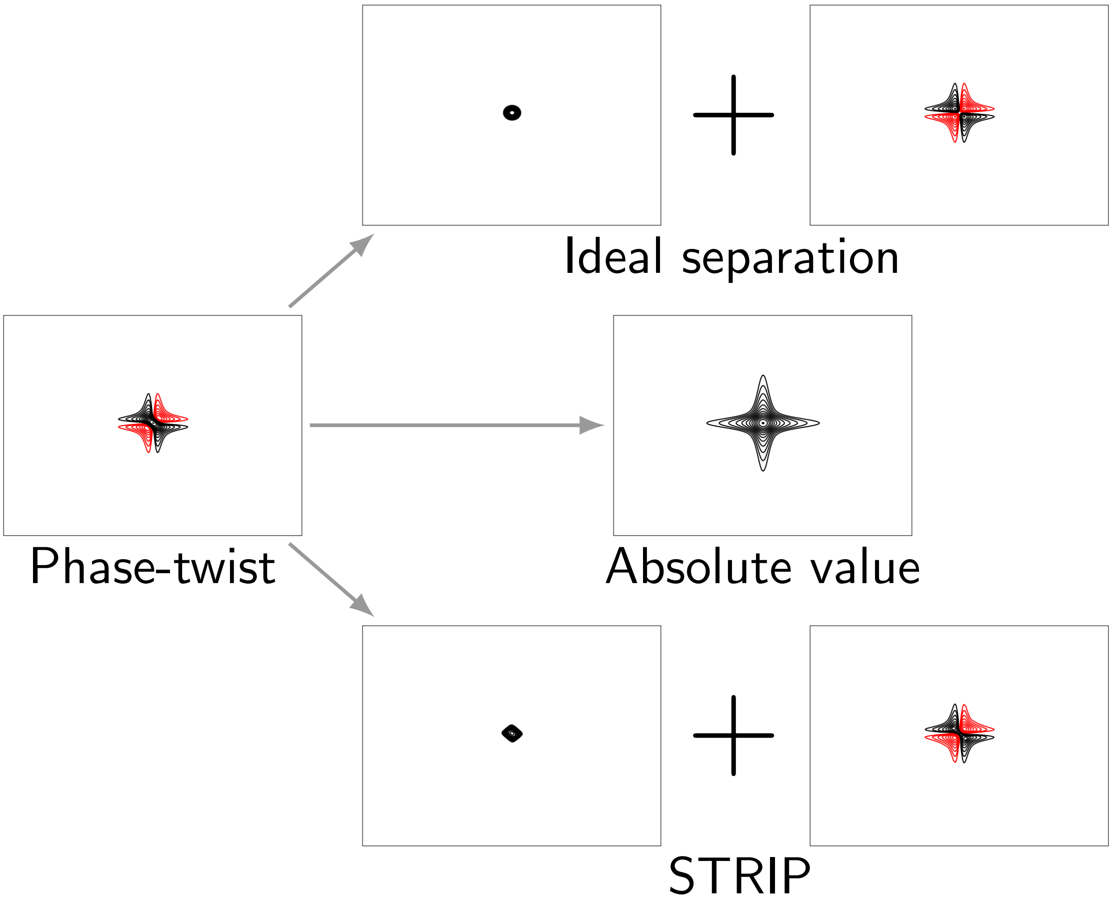

Figure 4: The result of applying a resolution enhancing window function to the time-domain data before Fourier transformation. To aid comparison, the spectral widths of the plots are identical to those used in figure 1.

4.1. Peakshapes

Results of applying STRIP to a single phase-twist peak are shown in figure 1. The output of STRIP appears marginally broader than the ideal pure absorption peak, but still much narrower than the absolute value peak. Note that the centre of that dispersive component that is generated by STRIP has black contours implying that maximum intensity of the absorptive component is less than the maximum intensity of the input phase-twist peak. The loss of intensity can be measured to be about 11%. This loss in maximum intensity is due to STRIP converging to a peak that is not pure double absorption as shown in figure 3. That figure also shows that improvements to peakshapes on each additional iteration tail off even after just a few iterations. Complete convergence may take many iterations, especially in the presence of significant levels of noise. It is expected that users of STRIP will choose a number of iterations that provides an acceptable result rather than one that converges completely. Any possibly significant improvements from additional iterations would come at the cost of a much larger computation time.

Both pure absorption peaks and the absorptive component produced by STRIP have a star shaped appearance. This is due to the Lorentzian shape produced from Fourier transformation of an exponentially decaying signal. When using the amplitude modulated experiments that provide pure absorption peaks it is common practice to make peaks more circular and narrower by applying a window function to the time-domain data. It follows that the output of STRIP will also typically require some resolution enhancement. Conceptually there are two ways that resolution enhancement could be applied to narrow the absorptive peak produced by STRIP . It could be applied during the processing of the phase-modulated data that generated the phase-twist peak, or it could be applied after STRIP has run. These will be called *Early* or *Late* enhancements. Late enhancement would involve an inverse Fourier transform, application of the window function and finally a Fourier transform to generate a spectrum. However, such a scheme is very unforgiving on peaks with underlying distortions. For example, although to the naked eye the output of STRIP seems relatively symmetrical, when late enhancement is applied the underlying distortions are amplified. Ironically, aggressive late enhancement creates positive and negative tails reminiscent of a phase-twist. On the other hand there are significant advantages with early resolution enhancement. This is because the desired solution to the STRIP method, the absorption component of the phase-twist peak, will have a width narrowed by the resolution enhancement. This will positively affect the outcome of STRIP by ensuring that any converged solution only has distortions under that narrower peak. For this reason, only early resolution enhancement is recommended to be used with STRIP.

Figure 4 shows the result of applying resolution enhancement to the peak shown in figure 1. As expected the relative sizes of all the peaks are shrunk relative to those without resolution enhancement. However, the size of the resolution enhanced absolute value peak is still several times larger than the pure absorption and STRIP peaks without resolution enhancement. Whereas the pure absorption peak becomes circular at this level of enhancement, the shape of the peak produced by STRIP retains a certain degree of squareness. The window functions that have been used have been chosen so as to convert exponentially decaying time-domain data to a more Gaussian shape. The window function applied is of the form $e^{(-Lt-Gt^2)}$. The values of L and G used for figure 4 were -1.0 and 0.6, for figure 5 -0.2 and 0.8 and for figure 7 -0.6 and 0.18 respectively. The same parameters were used in both dimensions. The T_2 relaxation time used in all simulations was 1 second, apart from in figure 5 where a random value between 1 and 5 seconds was used for different peaks.

4.2. Multiple Peaks

So far only a single peak has been considered. Applying STRIP to a spectrum containing many phase-twist peaks will simultaneously improve all of the peaks. The execution time of each iteration is independent of the complexity of the input spectrum. It has already been shown that residual components that cannot be removed by the iterative method are hidden beneath the output peak and give rise to some peak distortion. It is interesting to consider what happens when there are multiple peaks. It is perhaps possible that when two peaks are close together that residual components from one peak are hidden under another. This would cause peakshape distortions that depend on the proximity of nearby peaks. Figure 5 shows a congested area of peaks that has been processed in a variety of ways. The resolution enhancement window used was identical for all three plots. It is immediately clear that STRIP gives a much clearer spectrum than the absolute value display. However, its output peakshapes were not quite as uniform as those in the idealised amplitude modulated data. As in the case of the isolated peak, the peaks have slightly rectangular shape. There is one section of the plots where two peaks are overlapping. This degree of proximity has caused some additional minor distortion to these two peaks.

4.3. Noise

Information from NMR experiments is derived from the modulated signal that is generated and recorded. An aspect of these experiments is that this signal is always accompanied by a certain amount of noise. This noise is of little interest, but its presence limits the sensitivity of the particular experiment. It follows that when designing a new method the ratio of signal-to-noise is always of interest.

It has already been described that due to convergence in the STRIP method occurring before pure absorption peaks are generated, there is some residual impurity in the peakshapes and the intensity of the peak is roughly 11% lower than might otherwise be expected. To get the full picture it is necessary to see how the amplitude of the noise is affected too. Figure 6 shows a comparison between levels of noise according to how the data has been processed. The original data was a $1024x1024$ complex matrix of normally distributed random values. This was zero-filled once and Fourier transformed to generate a $2048x2048$ spectrum. This untreated noise spectrum, which would be the noise accompanying a phase-twist peak, is the displayed in the top left of figure 6. This spectrum contains noise that conforms to Gaussian noise statistics, with a zero mean and equal standard deviations of positive and negative values. Taking the absolute value of the noise spectrum produces a matrix of real positive values. The distribution of values in such a noise spectrum has a Rayleigh distribution. The expectation of a value from this distribution is $\sigma\sqrt{\frac{\pi}{2}}$ where σ is the standard deviation of the Gaussian noise in the unmodified spectrum. The mean of a large number of samples from this distribution will be this expectation. A signal displayed in absolute value mode will be expected to be raised by the mean of the noise that lies beneath it. The treatment of absolute value spectra is typically to perform a baseline

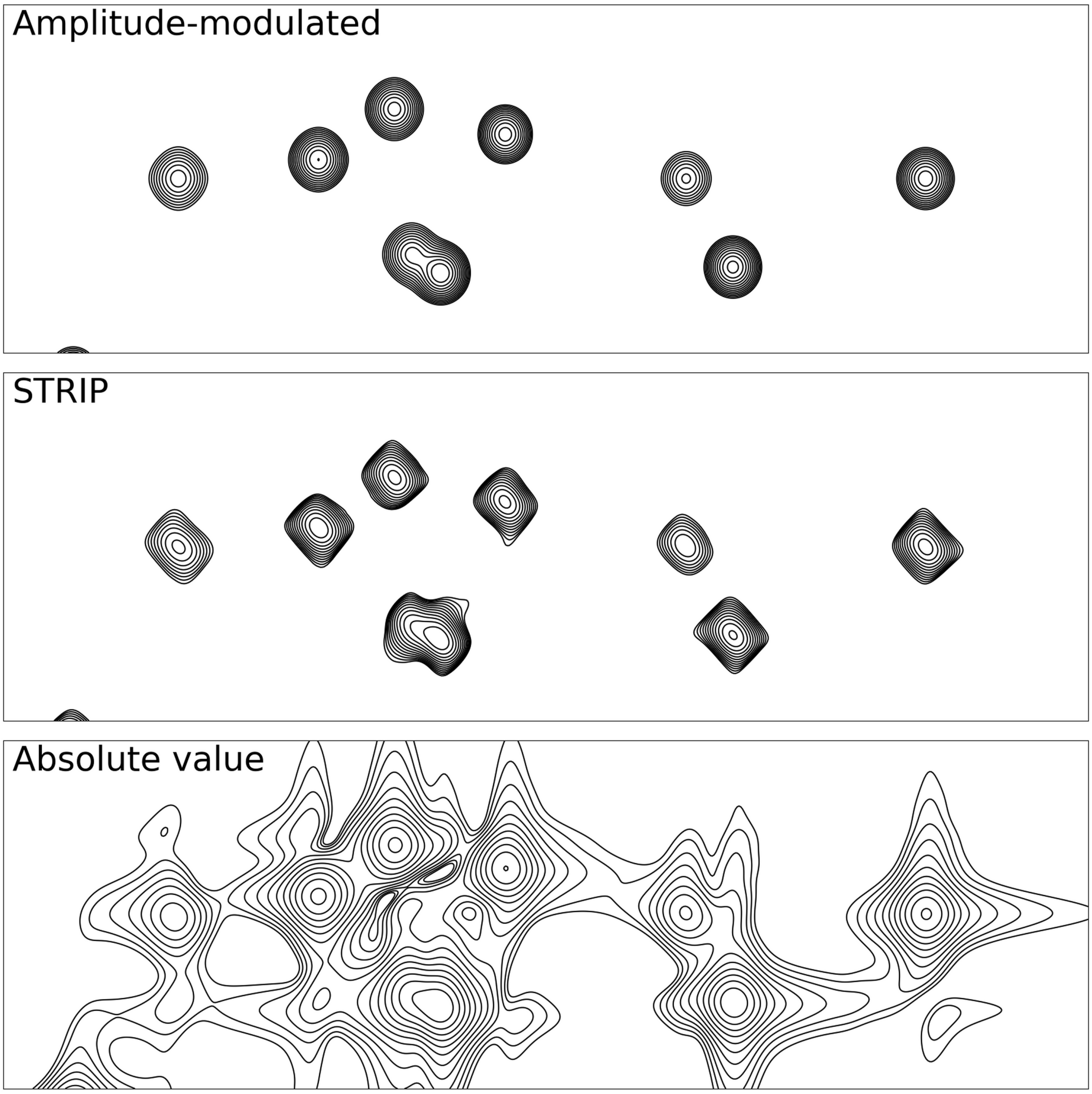

Figure 5: A spectrum was simulated containing a large number of peaks with random frequencies, intensities and relaxation times. The plots above zoom into the most congested area of the spectrum. In the top plot the simulation created amplitude modulated data. That data was resolution enhanced before Fourier transformation giving rise to circular peaks. For the lower two plots the simulation created phase-modulated data and this was resolution enhanced using the same window function as in the top plot. The phase-twist data in the middle plot was processed with STRIP and in the bottom plot displayed as an absolute value spectrum.

correction so that the expectation of a noise sample is zero. Note that once this transformation has been performed the shifted noise is not Gaussian. In common with Gaussian noise it now has a zero mean, however, its distribution is skewed to have a longer tail for positive values.

The distribution of noise produced after STRIP has been applied is even more highly skewed. In common with an absolute value spectrum, STRIP ensures that any NMR signals of interest are positive. It follows that the only noise that will be of concern is positive noise. Table 1 shows the relative levels of noise after applying different processing methods. Simply taking the standard deviation is flattering to the results of applying the STRIP method. For example, it suggests that noise in the untreated spectrum has amplitude over 70% greater. However, the untreated spectrum has Gaussian noise, whereas the noise spectrum produced by STRIP is highly skewed. To make a more realistic comparison a different measure is suggested that handles this skewness issue. The noise spectrum is split into two arrays

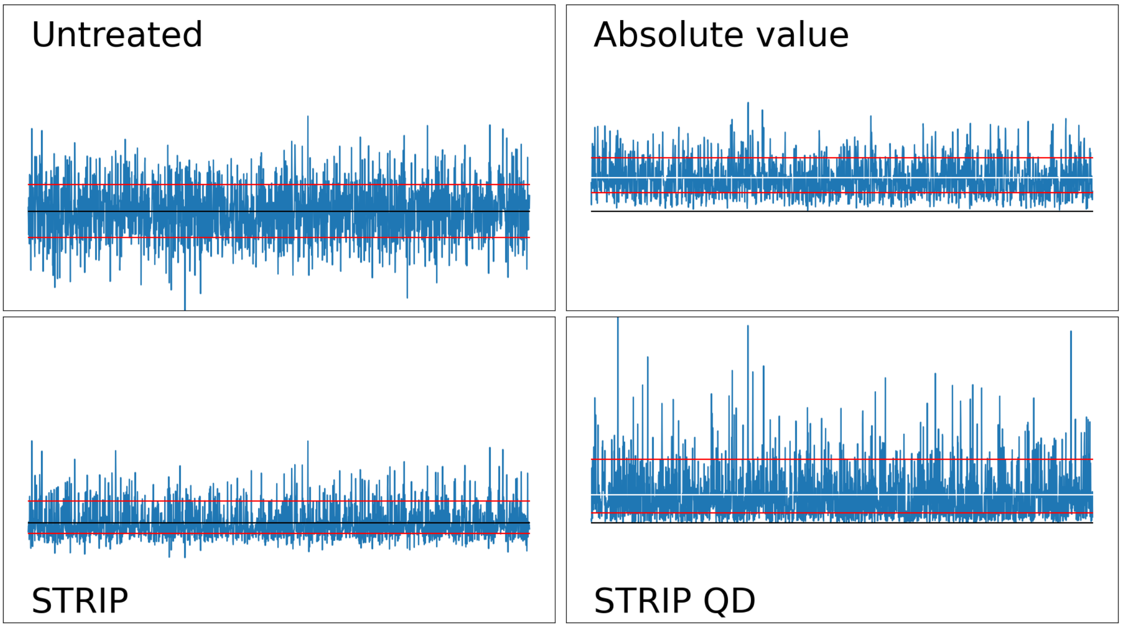

Figure 6: A view of how the background noise responds to different types of processing. A matrix containing noise was constructed, zero-filled, and then converted to a spectrum using a two-dimensional Fourier transform before being processed as indicated. The plots show a random trace from the 2D noise spectrum. The white line shows the mean of the noise, whereas the red lines show the separate standard deviations of samples above or below the mean. There is no white line visible in the plots on the left because the means of the noise in these plots are zero. A phase-twist peak would be accompanied by untreated noise. Strip QD refers to the quick-and-dirty method shown in listing 2.

	Untreated	Absolute Value	STRIP	STRIP QD
Std Deviation	1.716	1.124	1.000	1.685
Std Deviation above Mean	1.211	0.901	1.000	1.635

Table 1: A relative comparison of noise according to processing method. The standard deviations are shown relative to the value resulting from applying STRIP. Untreated noise is that in the real part of the initial complex spectrum. STRIP QD refers to the quick-and-dirty method shown in listing 2.

containing those samples that are either above or below the mean respectively. The only array of interest contains the samples above the mean and the standard deviation of these is used for the comparison shown in the second row of Table 1. Using this more relevant measure the relative noise advantage of STRIP falls against the untreated noise spectrum to just 21% and against the absolute value noise spectrum there is actually now a disadvantage.

Earlier it was mentioned that the STRIP method converged when taking the absolute value of a spectrum no longer had an impact. This would be the case when a spectrum is entirely positive. It is clear from the noise plots that this is not the full story, otherwise the noise in the plot showing the effect of STRIP would be entirely positive. It is believed that the reason for negative noise is that the input spectrum has negative noise too. A requirement for STRIP is that the input spectrum contains as signal positive phase-twist peaks, where the unwanted dispersive component is mixed with a positive pure absorption peak. The noise in the input spectrum contains negative phase-twist components which will not converge in the same way when processed with STRIP.

5. Discussion

There are large advantages to acquiring multi-dimensional NMR spectra in a way that produces amplitude-modulated data that provide absorption peaks[17]. It is the sensible choice wherever it is possible or feasible to acquire data in this way. It has already been mentioned that STRIP does not produce pure absorption peaks. An apt description would be that STRIP produces mildly contaminated absorption peaks. However, there are some experiments where it is just not possible or feasible to do anything else than acquire phase-modulated data. These are the ones where calculating absolute value spectra is the normal method of processing. It is for the processing of these types of experiments that the STRIP method is targeted. STRIP is extremely simple to implement and requires no data-specific parameters to be set (such as the number of expected peaks or the expected intensities and lineshapes of each peak). In common with Fourier transformation, the speed of execution of each iteration is independent of the complexity of the data that is being processed. The total speed of execution will only depend on the choice of the number of iterations to perform. STRIP is therefore entirely suitable to be embedded within standard NMR processing environments to provide an additional display mode to routinely complement or replace absolute value display.

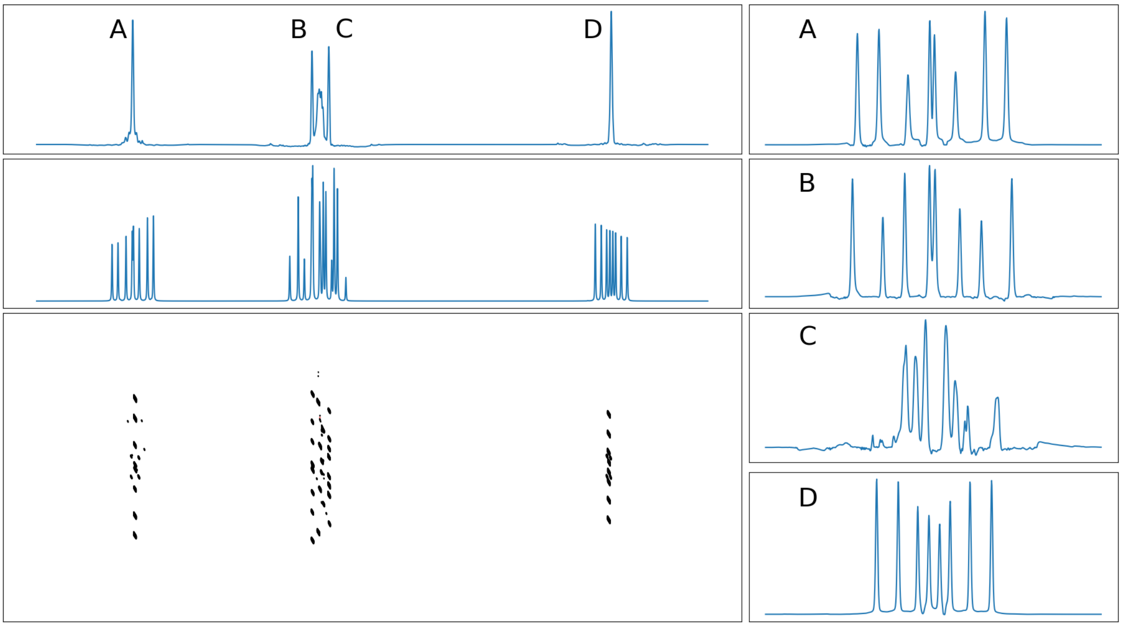

Figure 7: A J-Spectrum of a 4 spin system. The raw time-domain data was simulated using the GAMMA C++ library[15]. The spectrum was processed with STRIP and then sheared so that the projection along the F_1 axis gives a pure shift spectrum. Strong coupling effects give rise to peaks that are not aligned with the the positions representing the chemical shifts of each nucleus. These extra peaks are mainly between B and C, but close inspection shows them to be also close to A and D.

A prime example an experiment where applying STRIP would prove to be useful is two-dimensional J-Spectroscopy[18]. Potentially a highly useful experiment that can separate the effects of chemical shifts and scalar couplings, but blighted by phase-twist peakshapes. In figure 7 a simulated J-spectrum has been treated with STRIP as an example of what can be done. The spectrum is derived from time-domain data generated from a 4 spin system using the Gamma C++ library[15]. The large number of smaller peaks surrounding nuclei B and C are due to strong coupling. Note that negative strong coupling peaks will be distorted by STRIP. Modifying the effective free precession Hamilton within Gamma to treat scalar interactions as weak couplings makes these disappear. For real applications the J-Spectroscopy pulse sequence can be modified to aid elimination of these peaks[19].

The acquisition of phase-modulated data is still prevalent for magnetic resonance imaging(MRI) applications. These applications may benefit from the resolution enhancement that the new method can offer. An MRI image can be considered to be a densely packed spectrum consisting of many overlapping peaks. This is the sort of spectrum for which the STRIP method may converge early. However, even a single pass of the STRIP algorithm should improve resolution significantly. The application of this method to MRI images will be treated in a later paper.

Supplementary material

Source code for an implementation of the algorithm in Python is included within the text of the article. Source code for an implementation that uses CuPy and also for the scripts that used STRIP to generate the figures shown in this article can be found at https://github.com/C-J-Bauer/STRIP.

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